Stat 266 – Unit 6 – Video Set 1 – Class Examples

Summary:

Method of Distribution Functions

Note: We will only consider continuous random variables for this method.

Consider $U = h(Y_1, \ldots, Y_n)$ to be a function of the continuous random variables $Y_1, \ldots, Y_n$, which have joint density $f(y_1, \ldots, y_n)$.

- Find the region in the $(y_1, \ldots, y_n)$ space where $U \le u$.

- Find $F_U(u) = P(U \le u) = P(h(Y_1, \ldots, Y_n) \le u) = \int \cdots \int_{U \le u} f(y_1, \ldots, y_n) dy_1 \cdots dy_n$.

- Then, the density of U is $f_U(u) = \dfrac{d}{du} F_U(u)$.

Example A:

The amount of time waiting to be served, Y, is described by the following density

$$f(y)=\begin{cases}\frac{1}{\theta}e^{-y/\theta} & , \quad y>0 \\ 0 & , \quad else\end{cases}.$$

a) What is the probability distribution of $U=aY+b$, where $a>0$?

$$P(U\le u) = P(aY+b\le u) = P(Y\le \frac{u-b}{a}).$$

If $\frac{u-b}{a}<0$, that is, $u<b$, $P(Y<\frac{u-b}{a})$

$$=\int_{-\infty}^{\frac{u-b}{a}} f_Y(y)\,dy \quad \text{by } f_Y(y)\equiv 0 \text{ since } y\in(-\infty, \underbrace{\tfrac{u-b}{a}}_{<0})$$

If $\frac{u-b}{a}\ge 0$, that is, $u\ge b$.

$$P(Y\le \frac{u-b}{a}) = \int_{-\infty}^{\frac{u-b}{a}} f_Y(y)dy = \int_0^{\frac{u-b}{a}} \frac{1}{\theta}e^{-y/\theta}dy = -e^{-\frac{y}{\theta}}\Big|_0^{\frac{u-b}{a}}$$

$$= 1-e^{-\frac{u-b}{\theta a}} \quad \text{take derivative: } \int \frac{1}{\theta a}e^{-\frac{u-b}{\theta a}}. \quad b\le u$$

0 elsewhere.

b) What is the probability distribution of $U = Y^a$, where $a > 0$?

$$P(U \le u) = P(Y^a \le u) = P(a \ln Y \le \ln u) = P(\ln Y \le \tfrac{1}{a} \ln u).$$

$$= P(Y \le e^{\frac{1}{a}\ln u}) = P(Y \le u^{\frac{1}{a}}).$$

If $u^{\frac{1}{a}} < 0 \Rightarrow u < 0$ $\quad P(Y \le u^{\frac{1}{a}}) = 0.$

If $u^{\frac{1}{a}} > 0. \Rightarrow u > 0.$ $\quad P(Y \le u^{\frac{1}{a}}) = \int_{0}^{u^{\frac{1}{a}}} \frac{1}{\theta} e^{-\frac{y}{\theta}} dy$

$$= -e^{-\frac{y}{\theta}} \Big|_{0}^{u^{\frac{1}{a}}} = 1 - e^{-\frac{u^{\frac{1}{a}}}{\theta}}$$

Take deriviative: $\quad f_U(u) = \begin{cases} \frac{1}{\theta} u^{\frac{1}{a}-1} e^{-\frac{u^{\frac{1}{a}}}{\theta}} & u > 0. \\ 0 & u \le 0 \end{cases}$

Example B : If Y is a continuous random variable with density function

$$f_Y(y) = \begin{cases} \dfrac{3y^2}{16} & , \quad -2 < y < 2 \\ 0 & , \quad else \end{cases},$$

what is the density function of $U = 1 - 6Y^2$?

$$P(U \le u) = P(1 - 6Y^2 \le u) = P(1 - u \le 6Y^2).$$

$P\left(\frac{1-u}{6} \le Y^2\right)$. $u > 1$, $\frac{1-u}{6} < 0$. $P\left(\frac{1-u}{6} \le Y^2\right) = 1$.

과 $u \le 1$, $P\left(\frac{1-u}{6} \le Y^2\right) = P\left(Y \le -\sqrt{\frac{1-u}{6}}\right) +$

$$P\left(Y \ge \sqrt{\frac{1-u}{6}}\right) = \int_{-2}^{-\sqrt{\frac{1-u}{6}}} \frac{3y^2}{16}\,dy + \int_{\sqrt{\frac{1-u}{6}}}^{2} \frac{3y^2}{16}\,dy$$

$$= \frac{1}{16}y^3\Big|_{-2}^{-\sqrt{\frac{1-u}{6}}} + \frac{1}{16}y^3\Big|_{\sqrt{\frac{1-u}{6}}}^{2} = 1 - \frac{1}{8}\left(\sqrt{\frac{1-u}{6}}\right)^3 = 1 - \frac{1}{8}\left(\frac{1-u}{6}\right)^{\frac{3}{2}}$$

Take derivative: $-\frac{1}{8} \cdot \frac{3}{2} \cdot \left(\frac{1-u}{6}\right)^{\frac{1}{2}} \cdot -\frac{1}{6} = \frac{1}{32}\sqrt{\frac{1-u}{6}}$.

Example C:

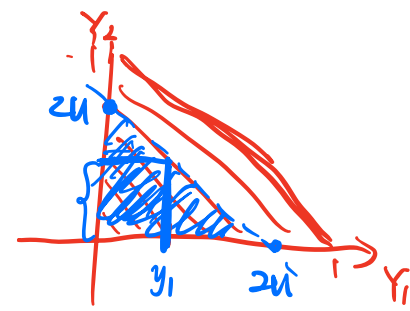
The life (in hundreds of hours) of a type of battery running a particular machine follows an *Exponential* distribution with mean 1. All batteries are independent. Let X denote the life of battery 1, and Y denote the life of battery 2.

a) What is the probability density of the average life of two of these batteries?

$$\int_0^{+\infty} y \frac{1}{\theta} e^{-\frac{y}{\theta}}\, dy = -y e^{-\frac{y}{\theta}} \Big|_0^{+\infty} - \int_0^{+\infty} e^{-\frac{y}{\theta}}\, dy$$

$$= \int_0^{+\infty} e^{-\frac{y}{\theta}}\, dy = -\theta e^{-\frac{y}{\theta}} \Big|_0^{+\infty} = \theta.$$

Since $E(Y)=1 \Rightarrow \theta=1$ $\quad f_Y(y) = \begin{cases} e^{-y} & y>0 \\ 0 & \text{otherwise.} \end{cases}$ the joint density function



is $f_{Y_1,Y_2}(y_1,y_2) = \begin{cases} e^{-(y_1+y_2)} & y_1>0 \quad y_2>0 \\ 0 & \text{otherwise.} \end{cases}$ $\qquad U = \frac{Y_1+Y_2}{2}.$

$$P(U \le u) = P\left(\frac{Y_1+Y_2}{2} \le u\right)$$

$P(Y_1+Y_2 \le 2u).$

$u<0,\ P(Y_1+Y_2 \le 2u) = 0 \qquad u>0.$

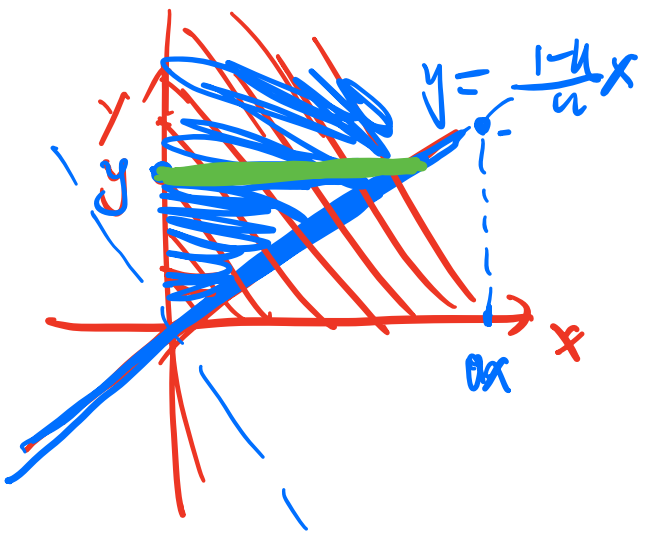
$$\int_0^u \int_0^{2u-y_1} e^{-(y_1+y_2)} \cdot dy_2 \cdot dy_1 = \int_0^{2u} \left(-e^{-(y_1+y_2)}\right)\Big|_0^{2u-y_1} dy_1 = \int_0^{2u} \left(-e^{-y_1-2u+y_1} + e^{-y_1}\right) dy_1$$

$$= -e^{-y_1} - y_1 e^{-2u} \Big|_0^{2u} = 1 - e^{-2u} - 2u e^{-2u}$$ take derivative

$$f_U = \begin{cases} 4u e^{-2u} & u > 0 \\ 0 & \text{otherwise} \end{cases}$$

b) Consider two batteries. What is the probability density for the proportion of the total time the machine ran off battery 1? That is, what is the distribution of $U = \dfrac{X}{X+Y}$?

$$P(U \le u) = P\left(\frac{X}{X+Y} \le u\right) = P(X \le u(X+Y)) = P(0 \le (u-1)X + uY).$$



Consider $(u-1)X + uY = 0$.

If $0 < u < 1$, $y = \frac{1-u}{u}x$ intersect with red area.

fix $Y = y$ as a constant. $X(u-1) \ge -uy$.

$$x \le \frac{-uy}{u-1}.$$

$$\int_0^{+\infty} \int_0^{\frac{-uy}{u-1}} e^{-(x+y)}\, dx \cdot dy$$

$$= \int_0^{+\infty} -e^{-(x+y)} \Big|_0^{\frac{-uy}{u-1}} dy = \int_0^{+\infty} e^{-y} - e^{-(\frac{-u}{u-1}+1)y}\, dy.$$

$$= \int_0^{+\infty} e^{-y} - e^{\frac{y}{u-1}}\, dy$$

$= -e^{-y}\cdot 0 - (u-1)e^{\frac{y}{u-1}}\Big|_0^{+\infty} = 1 + (u-1) = u$ take derivative

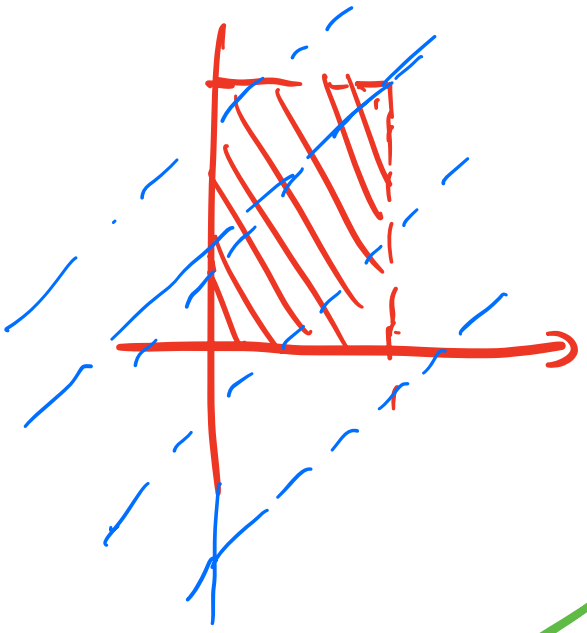
$u' = 1$

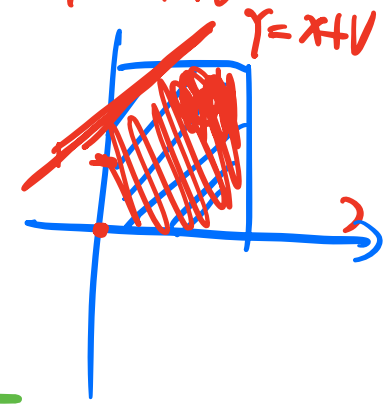
Example D : The joint density of X and Y is given as

$$f(x,y) = \begin{cases} \frac{1}{2} \ , & 0 < y < 2, 0 < x < 1, \\ 0 \ , & else \end{cases}$$

Find the probability density function of $V = Y - X$.

$P(V \le v) = P(Y - X \le v) = P(Y \le X + v)$. consider $Y = X + V$.





If $1 < V < 2$

$P(Y \le X+v) = 1 - \underbrace{P(Y \ge X+v)}_{\text{compute this one}}$.

$$\int_{\text{Green}} f(x,y)\,dx\,dy = \int_V^2 \int_0^{y-V} \frac{1}{2}\,dx\,dy.$$

$$= \int_V^2 \frac{1}{2}x\Big|_0^{y-V} dy = \int_V^2 \frac{1}{2}(y-V)\cdot dy$$

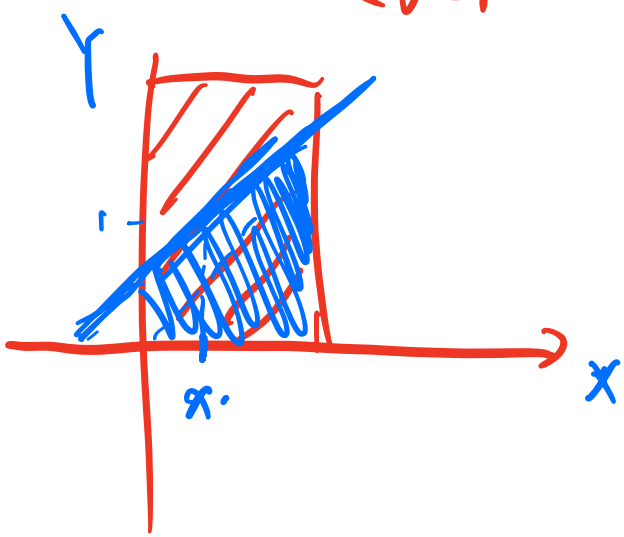
$$= \frac{1}{4}y^2 - \frac{1}{2}vy \Big|_v^2 = 1 - v - \frac{v^2}{4} + \frac{1}{2}v^2$$

$$= 1 - v + \frac{v^2}{4}$$

$$P(Y \le x+v) = 1 - \left(1 - v + \frac{v^2}{4}\right) = v - \frac{v^2}{4} \qquad 1 \le v \le 2$$

total deriviative : $1 - \frac{v}{2}$ $\qquad 1 \le v \le 2$

$0 < v < 1$



$$\int_0^1 \int_0^{x+v} \frac{1}{2}\, dy\, dx$$

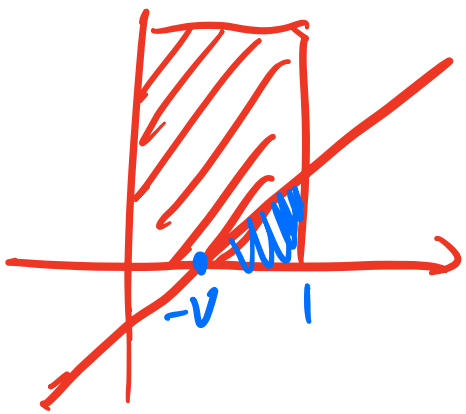
$$= \int_0^1 \frac{1}{2} y \Big|_0^{x+v} dx$$

$$= \int_0^1 \frac{1}{2}(x+v)\, dx = \frac{1}{4}x^2 + \frac{1}{2}xv \Big|_0^1$$

$$= \frac{1}{4} + \frac{1}{2}v \qquad 0 < v < 1.$$

take derivative $\frac{1}{2}$. for $0<v<1$.

对于 $-1<v<0$.



$$\int_v^1 \int_0^{x+v} \frac{1}{2}\, dy\, dx$$

$$= \int_{-v}^1 \frac{1}{2}(x+v)\, dx = \frac{1}{4}x^2 + \frac{1}{2}xv \Big|_{-v}^1$$

$$= \frac{1}{4} + \frac{1}{2}v - \frac{1}{4}v^2 + \frac{1}{2}v^2$$

$$= \frac{1}{4} + \frac{1}{2}v + \frac{1}{4}v^2.$$

derivative $= \frac{1}{2} + \frac{1}{2}v.$

$$f_V(v) = \begin{cases} 1 - \frac{1}{2}v & 1 < v < 2 \\ \frac{1}{2} & 0 < v \leq 1 \\ \frac{1}{2} + \frac{1}{2}v & -1 < v < 0 \\ 0 & \text{otherwise.} \end{cases}$$